

SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares total site energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. The construction cost data is from 2026 RSMeans Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_NY_Buffalo-Greater.Buffalo.Intl.AP.725280_TMY3.epw
- Building Name: SmallOffice
- Building Location: Buffalo, NY
- Total Floor Area: 511 m²

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall + roof + window + door	Wall upgrade: • Exterior wall insulation: Blown Fiberglass, target R-20.4 Roof upgrade: • Roof insulation: Blown Fiberglass, target R-34.5 Window upgrade: • 2-pane replacement (skipped)

		- Weatherstrip application (skipped: no OperableWindow) - Glazing film application (skipped) - Caulking application (skipped) - Window frame replacement (skipped) Door upgrade: - Bottom seal: silicone adhesive smoke gasket - Top/side seal: silicone adhesive smoke gasket
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Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	440.31	388.03	52.28	11.87%

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison

Baseline		\$18,286
Scenario 1		\$17,052

■ Baseline ■ Scenario 1

Max saving: **\$-1,234 (-6.75%)**

Total Site Energy Saving (Scenario - Baseline) (GJ/yr) Comparison

Baseline		440
Scenario 1		388

■ Baseline ■ Scenario 1

Max saving: **52 GJ (12%)**

Min Retrofit Construction Cost

\$11,631

Scenario 1

Lowest Retrofit Cost Payback Period

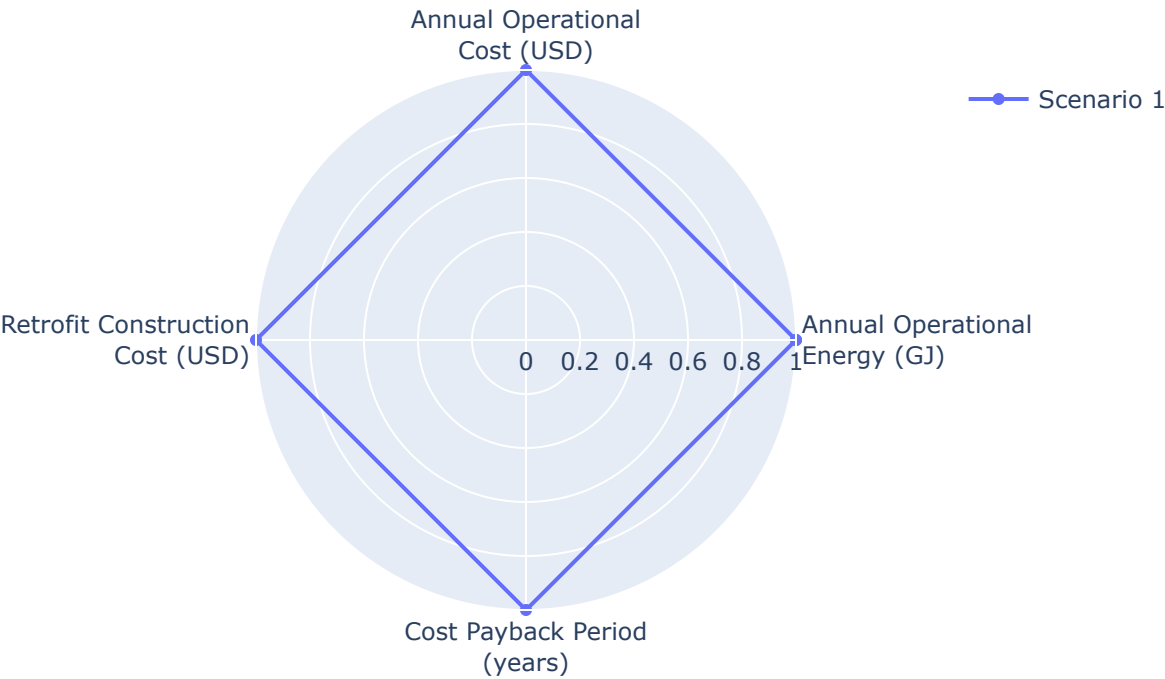
9.43 years

Scenario 1

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization

Parametric Scenario Comparison (Normalized Spider Chart)



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).

Scenario 1 9.43 yrs

Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

Scenario	Component	Material name	Volume (m³)	Area (m²)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m³)	Product Lifetime (years)
Scenario 1	Wall insulation	Blown Fiberglass	17	282	N/A	N/A	0.03	30	30
Scenario 1	Roof insulation	Blown Fiberglass	144	599	0.24	N/A	0.03	30	30

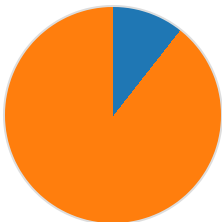
Scenario 1	Door bottom sealing	silicone adhesive smoke gasket	N/A	N/A	N/A	3.66	N/A	N/A	15
Scenario 1	Door top/side sealing	silicone adhesive smoke gasket	N/A	N/A	N/A	16	N/A	N/A	15

Result Summary

Per-scenario breakdown: left chart shows retrofit construction cost contribution by measure, right chart shows total site energy by fuel type (electricity, natural gas, and other). Water is excluded from energy breakdown.

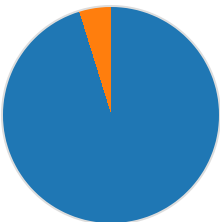
Scenario 1

Cost breakdown by retrofit measure



■ Wall: \$1,230 (10.6%)
■ Roof: \$10,401 (89.4%)
Total: \$11,631

Total Site Energy Breakdown (GJ)



■ Electricity: 369.41 GJ (95.2%)
■ Natural Gas: 18.62 GJ (4.8%)
Total: 388.03 GJ

Scenario	Retrofit Construction Cost (USD)	Annual Operational Cost (USD)	Total Site Energy (GJ)
Baseline	\$0.00	\$18,286	440.31
Scenario 1	\$11,631	\$17,052	388.03

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